

Element H:

Cardboard Product:



The prototype that we built was made entirely out of cardboard and popsicle sticks and held together by hot glue and tape. We decided to keep the materials that we built this prototype with at a minimum to replicate the important factor of a lightweight, simple box. Many pieces of cardboard we cut came from extra, unused boxes from the engineering room, but those pieces happened not to be sturdy enough to successfully replicate our design. To fix this problem, we added popsicle sticks on the creases of the cardboard where durability was lacking. The dimensions of the box are 16" width by 21" length with 12" height. Along with this, the width of all of these pieces was 3.25".

As stated, the simplicity and weight of this box were our highest priorities. The zookeepers required that the box be carried by 2 people, which means it is a relatively lightweight box because of the heavier weight of the Komodo dragon. Because the simplicity of

the box is also a high priority, our prototype has a simple frame, with the only complication in our prototype being the durability.

The design of the cage was tested for reliability. The test that the group conducted was putting a dog into the cardboard design. Although this is not a proficient test, it was helpful in a sense to determine if a live animal could be used. The testing showed that the dog acted well in the cage, however, it was not durable enough to survive its one-time use. We suspected that this would happen because of the flimsy cardboard material, but it was a good experiment to see if the group liked the general design.

The zoo experts have made it clear that it was vital that the materials used could handle the force of the Komodo dragon. Keeping this in mind, our final design will be made sure to represent the sturdiness of the design.

After we were done with our part, each of the other 2 teams were able to use this same prototype, and they added their portions to it. This was all discussed during our weekly team meetings.

For evaluation, the prototype was proficient in dimensions and ease of movement. However, if we were to spend more time on this prototype and improve it, then we would improve durability to survive against the resistance of the dragon.

Metal Product:

HANDLES TESTING PLAN

1. Test the strength, comfort, and ease of movement of the handles:

- a. Attach two of the wood slabs with handles attached to them with another piece of wood with screws to see how the handle would look when attracted to a frame



- b. Have two people hold this contraption, with one person per handle
- c. Begin adding known-mass weights until the handles fail or 120 lbs (~ 55 kgs) is achieved. For each addition of mass, lift the contraption and allow for 30 seconds before adding another mass.
- d. If the handles fail, record the last successful mass and the point of failure on the handles.
- Other tests for the handles will be if they can smoothly fold in and out.
Check for squeaks and ease
 - A group of people will hold the handles and rate the comfortability on a scale of 1 to 5

PLASTIC TESTING

Floors ($\frac{1}{4}$ " thick)

1. Add increasingly heavy weighted plates onto the plastic to represent a Komodo dragon laying on it
2. After certain amounts of time, remove the plates and observe the plastic
3. Make sure that the plastic can withstand the weight with little to no stress/cracks

Walls (1/8" thick)

1. Support the plastic on all four sides
2. Using a baseball bat or other metal bar, whack the plastic across multiple points of the plastic, with each whack starting only 6" away from the wall with no swing back. This is to represent the dragon's tail and the limited range it has to swing.
3. Observe and record damage, including cracks, scratches, and dents.

BOX TESTING PLAN

1. Once the boxes have been put together, there are many tests that need to be run on them
 - a. First, we will test the weight of each box to make sure that they aren't too heavy, as the criteria is that it needs to be able to be held by two people.
 - b. Use a hammer to bang the plastic sheets on the walls and ceiling to mimic the tail of the Komodo dragon banging on the sides. Make sure that there aren't any cracks or stress marks on the plastic after the hammering is done.
 - c. Make sure that the boxes are able to stack on top of each other (doesn't matter if it is on the top or the sides)

Material Testing

It is important to be able to clean the cage after each use.

- Use a spray bottle with a liquid of choice, or dirt
- Ask peers to wipe the mess completely
- Obtain personal rating of how easy it was to clean

Locking mechanism

The safety of the zookeepers is a top priority, so keeping the dragon locked in the box is a criteria.

- Ask peers to get close enough to the gate where they can attempt to open it
- Log if the lock opened or if it upheld force and kept itself locked